



SMART TRAVEL GUIDER WEB APPLICATION FOR SRI LANKAN TOURISM

An Intelligent Full-Stack Platform for Automated Itinerary Planning, Cultural Exploration, and Service Booking

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Problem & Motivation

Independent tourists in Sri Lanka struggle with fragmented information and complex manual itinerary planning across disconnected platforms.

- **Real-World Problem:** Tourists waste over 15 hours coordinating disjointed hotels, transport providers, and tour guides manually.
- **Measurable Scale:** Over 1.4 million annual tourists face booking friction, causing 25% itinerary dropouts and \$40M lost revenue.
- **Current Gap:** Existing ad-hoc booking workflows lack real-time local service integration, dynamic

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Development Methodology

Adopted an Agile Scrum framework to enable rapid feature iteration and continuous user evaluation.

- **Model:** Agile Scrum — chosen to adapt continuously to tourist UX feedback and vendor requirement changes.
- **Phases:** Requirements Gathering → Schema Design → REST API Build → React UI Integration → System Testing → Deployment.
- **Sprint Structure:** 7 bi-weekly sprints (14 weeks total) managed using Jira and GitHub Projects.
- **Requirements Source:** Formulated from 45

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Key Features

- **Automated Itinerary Generator:** Create tailor-made daily plans based on budget, travel dates, and personal interests.
- **Integrated Service Bookings:** Reserve certified hotels, licensed tour guides, and private transport in a single checkout workflow.
- **AI Travel Assistant:** Obtain 24/7 intelligent cultural recommendations, route tips, and local guidelines via AI chat.
- **Multi-Role Management:** Vendor management for local suppliers and analytics dashboard for platform admins.